

CSE 4508

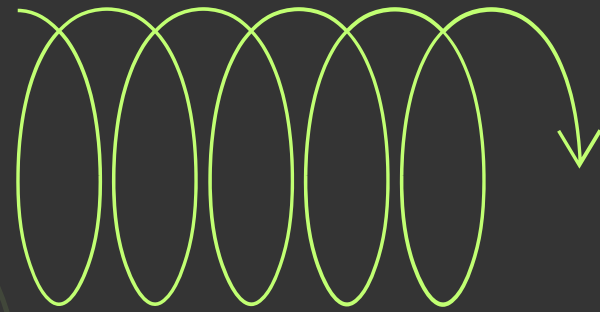


XOR OJ



The Team

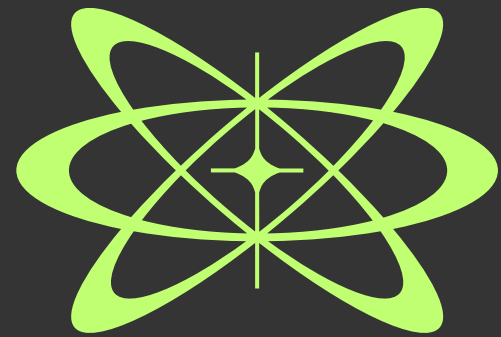
Kabir • **Mishkat** • **Nayeem** • **Sattyam**
220041109 **220041134** **220041139** **220041141**



Project Overview

XorOJ is a full-stack online judge system designed to host programming contests, accept and judge submissions, and provide real-time standings. The system will be able to support multiple programming languages and secure user authentication.





Key Features



User Portal

Registration, login, and progress tracking.

Problem Integration

Unified interface for problems from Codeforces, CodeChef, AtCoder, LightOJ, and more.

Contest System

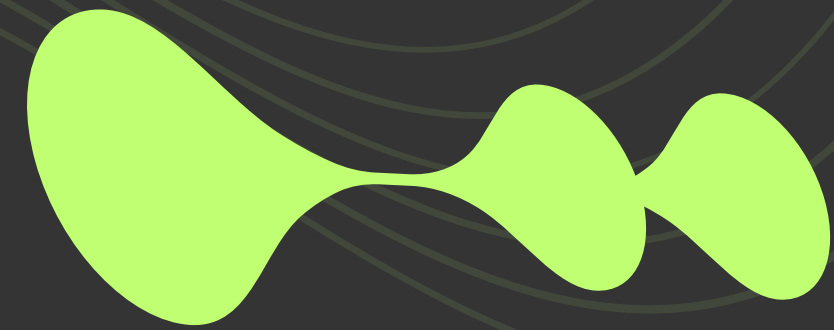
Creation and scheduling of diverse coding contests.

New Problems

Admin interface for adding custom coding challenges.

Online IDE

In-browser code editor for immediate compilation and testing



Tech Stack

DBMS Choice: PostgreSQL

Frontend : React + Vite

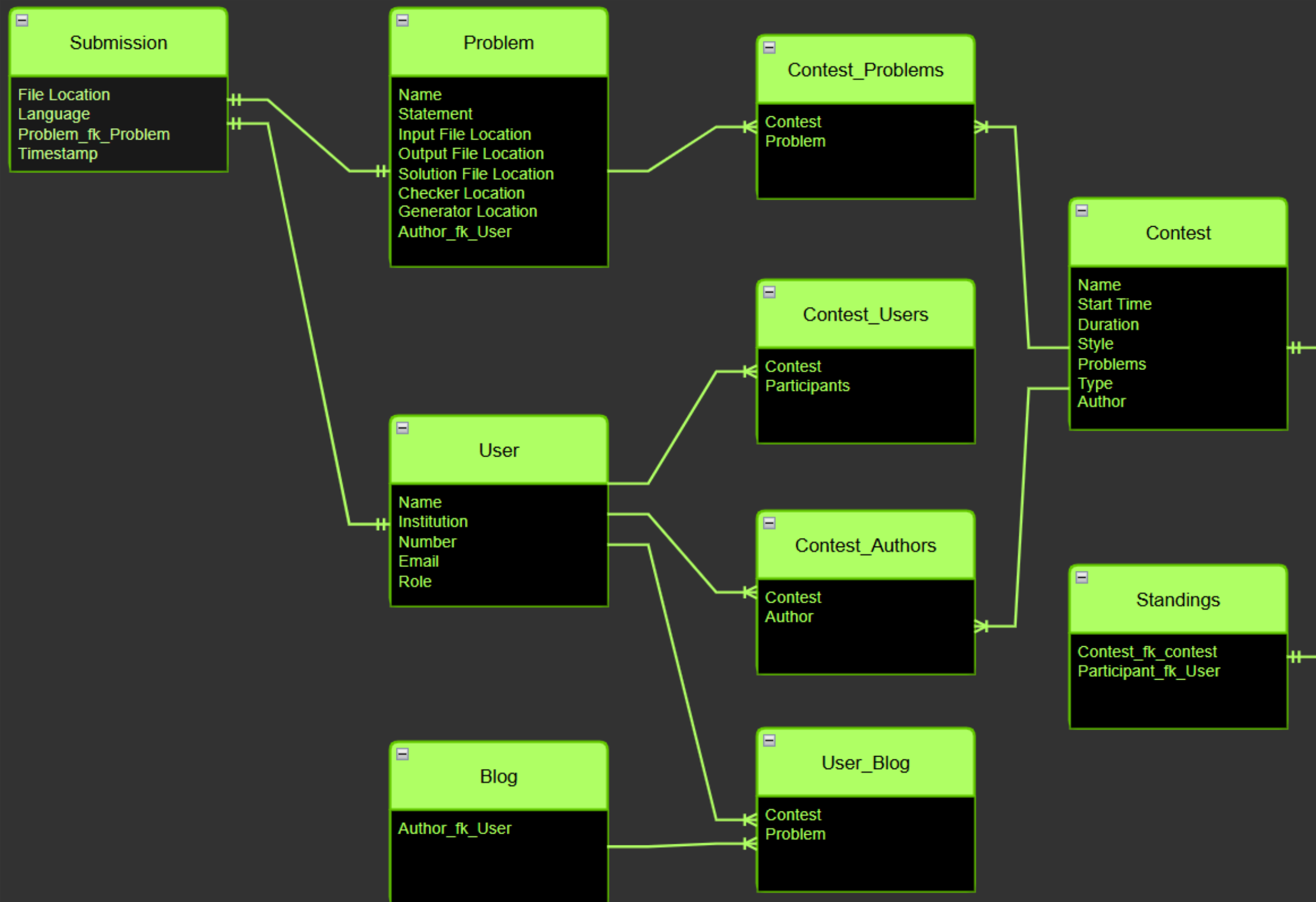
Backend : Spring Boot

Authentication: JWT

Version Control: Git + GitHub

Draft ER Diagram

R • D • B • M • S



Database Components

R • D • B • M • S

Stored Procedures

sp_AccessControl
sp_Submission
sp_UpdateContestPrb
sp_CreatePrb

Triggers

trg_UpdateStandings
trg_UpdateParticipantsList

DEMO

Scan for GitHub Link





THANK YOU